Elicit vs. Teach, Mechanistically: Results on the TinyStories-1B Twins

MARS V — Mechanistic Understanding of Elicitation vs. Teaching

Working results document, September 2026

Setup

Two identical TinyStories-1B twins (same architecture, same pretraining on children's stories, zero arithmetic) are fine-tuned on the same target task — signed 1–4-digit addition/subtraction in natural language, `What is the difference between 8 and 5784?` → `-5776` — with one byte-held protocol (LoRA r=512, α=32, lr 3.53×10^{-4}, batch 128, seed 316, eps/k convergence stop, frozen D_algo_bare, $n = 10^2 \dots 10^6$). The arms differ *only* in the parent checkpoint:

- **Teach twin**: the blank base (`evt-ts1b-base`). The capability is absent.
- **Elicit twin**: **TS1B-latent** (`evt-ts1b-op-bridge-mix`) — the blank base plus (i) a symbol-only arithmetic install (`23 + 45 = 68`, 1M examples, full FT) and (ii) an answer-free word↔symbol binding dose (rewriting `What is the sum of 621 and 5068?` ↔ `621 + 5068`, rehearsed 1:1 with already-seen symbol rows). Every installation is answer-free and leakage-controlled (Table 1). The capability is latent.

Construction step / check	value	control
symbol install: EM on `23 + 45 =`	0.67–0.73	blank: 0.00
corpus census: "sum" in 439M words	21×	"equals" 17, "minus" 57
binding dose: NL→op rewrite exact	0.484	blank + dose: 0.000 EM
sum-only dose on *difference* questions	writes "+"	per-word binding
direct NL exact match of the parent	**0.000**	= blank twin

Table 1: The latent parent is behaviorally identical to the blank twin on the target task before fine-tuning. Everything below is about what differs inside and what that difference does.

Measurement vocabulary. *Circuit* = top-32 of 528 nodes (512 attention heads + 16 MLP outputs) by attribution score (gradient of the clean/corrupt logit-diff × activation delta, 128–256 pairs); overlaps are Jaccard@32 against a chance floor of 0.031 and a per-map split-half ceiling. *Gradient strength* = the pre-clip global gradient norm $\|g_t\|$ logged at every update (runs' `gradstats.jsonl`); cumulative gradient mass $\sum_t \|g_t\|$. *Weight write* = ΔW = scaling $\cdot\, BA$ per LoRA module (exact); total write $\|\Delta W\|_F$, relative write $\|\Delta W\|/\|W\|$, travel $\|\Delta W_t\|_F$ from adapter snapshots.
Reading guide. The results answer four questions in order: does the *behavior* of learning differ (R1); is the resulting *machinery* the same (R2–R4); how does that machinery *come into being* during training (R5–R8); and what does the latent state *mean and permit* before any target training (R9–R11).

A. Behavior: the learning-curve signature

R1 The learning-curve signature flips.

Metric. Prequential EDL per label token (nats) and held-out exact match, as a function of dataset size n, identical protocol, parent varied.

Result. Figure 1, Table 2. Elicit: EDL strictly monotone, $4.53 \to 0.092$; EM 0.54 at n=3,162, 0.98 at 1M. Teach: EDL down–up–down with the hump peaking near n=215K; EM 0.093 at 1M. EM-0.5 threshold: $\sim 3 \times 10^3$ vs $> 10^6$ ($> 300\times$). A third arm (blank + format-only dose) keeps its hump.

Elicit vs. teach. Eliciting spends every example connecting an existing capability (diminishing cost from the first rung); teaching must build the capability, producing increasing returns mid-run. The flip is the paper's causal-intervention claim reproduced on one substrate, with a parent whose latency is itemised rather than assumed. The remaining results explain the flip mechanistically.

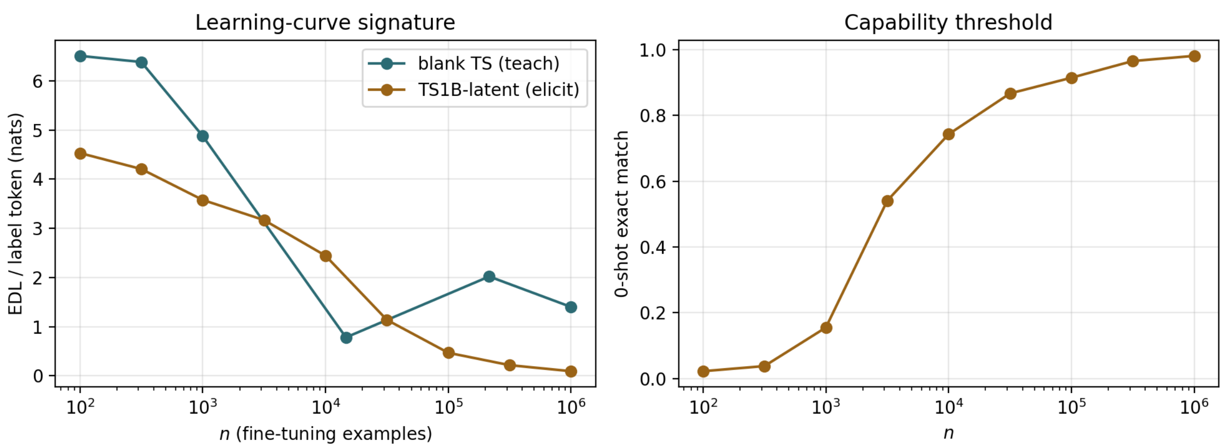

Figure 1: EDL/token (left) and 0-shot EM (right) vs n; teal = teach twin, gold = elicit twin.

n	elicit		teach	
	EDL/tok	EM	EDL/tok	EM
100	4.533	0.023	6.511	0.007
316	4.208	0.038	6.387	0.033
1,000	3.579	0.155	∼4.9	∼0
3,162	3.168	**0.541**	↓	∼0
10,000	2.444	0.743	min ∼0.8	∼0
31,623	1.137	0.867	↑	∼0
100,000	0.467	0.915	↑	∼0
316,228	0.215	0.966	peak ∼2.0 @215K	∼0
1,000,000	**0.092**	**0.981**	1.40	0.093

Table 2: Per-rung values (EDL nats/label token; EM on 1,024 held-out problems).

B. Machinery: is the circuit the same?

R2 The elicited circuit is the parent's installed circuit; the taught circuit matches nothing upstream.

Metric. Jaccard@32 between attribution circuits (chance 0.031; ceiling = split-half self-agreement of the map, 0.684 for the elicited child).

Result. Table 3. Installed engine ↔ elicited child: 0.455–0.524 across eval surfaces (≈70% of ceiling), top-4 nodes identical and identically ordered (`mlp:15,14,13,12`); the engine backbone is already fully shared at n=3K (child@3K ↔ child@1M 0.391, with only the input interface churning). Blank parent: no measurable circuit (logit-diff −0.33 even 16-shot; its map is noise), so the taught child has nothing to match.

Elicit vs. teach. Elicitation re-weights machinery that was already there; teaching constructs machinery where there was none. "Reuse" is measured, not inferred.

Comparison	J@32	chance	ceiling
installed engine ↔ elicited child (op / bridge surface)	0.455–0.488 / 0.524	0.031	0.684
elicited child @3K ↔ @1M	0.391	0.031	0.684
blank parent ↔ taught child	undefined (parent map is noise)	–	–
elicited child @1M ↔ taught child @1M	0.231 (Spearman −0.11)	0.031	0.684 / 0.561

Table 3: Circuit-membership overlaps. All maps pass the performing-regime guard.

R3 The parent's functional head roles survive elicitation intact; the taught child does not use them.

Metric. Desiderata Component Masking (Prakash et al. 2024), heads only: for each functional variable — operand a, operand b, operation (+ ↔ −) — learn the sparse set of attention heads whose counterfactual activations, patched into the clean run, flip the answer to the counterfactual one; compare role sets across models by Jaccard.

Result. Table 4. Parent (installed engine) vs elicited child on the op surface, with cf-flip accuracy equal to the full-counterfactual ceiling in every set: operand-a heads shared 29/33 (J 0.879), operand-b 27/30 (0.900), operation 21/28 (0.750); chance ≈ 0.05. The operand roles are essentially the whole of layer-0 attention (30/32 heads) — operand *reading* is a layer-0 function — and the operation role spans heads in layers 0–5. On the NL surface, the taught child's operand heads overlap the elicited child's at J 0.125/0.085 and the parent's at 0.147/0.091, and the mask cannot find a compact flipping set in it (cf-flip 0.02–0.04 vs a ceiling of 0.15).

Elicit vs. teach. Elicitation preserves *function*, not just membership: the same heads fetch operands and detect the operation before and after a million target examples — roles are even more stable than circuit membership (0.88 vs 0.455). The taught child does not carry operand information through those heads; whatever it uses is diffuse enough that no compact fetcher set is recoverable (a low-confidence characterisation — its NL first-token ceiling is only 0.15 — so "roles created" is supported in the weak form: not the parent's roles).

R4 Same task, same substrate, different machine — at every level.

Role	|parent|	|child|	shared	Jaccard	chance	layers
operand a	30	32	29	**0.879**	0.055	0
operand b	27	30	27	**0.900**	0.051	0
operation	25	24	21	**0.750**	0.041	0–5
elicited vs taught child (NL): operand a / b	36 / 30	9 / 21	5 / 4	0.125 / 0.085	0.02	–

Table 4: DCM head roles (heads only; cf-flip accuracy = ceiling in all elicit-side sets).

Metric. Layer distribution of the top-16 attribution nodes; J@32 and score Spearman between the two children; node- and edge-level change rates $\Delta S = 1 -$ Jaccard between them (nodes @32; edges @256, node→block edge attribution).

Result. Figure 2. Elicited child: mid/late MLP backbone (layers 5–15) plus the layer-7 attention trio — the installed engine's own profile. Taught child: eight layer-0 attention heads over raw digit tokens feeding a few late MLPs. Overlap 0.231, score correlation −0.11; $\Delta S_{\text{node}} = 0.769$, $\Delta S_{\text{edge}} = 0.739$, both far above the noise floors (0.32 / 0.61).

Elicit vs. teach. Because both children share the identical pretrained substrate, the mechanism difference is caused by the learning regime alone (the cross-model version of this finding had a pretraining confound). And the difference is total — membership and wiring alike — so these are two machines, not one machine re-routed.

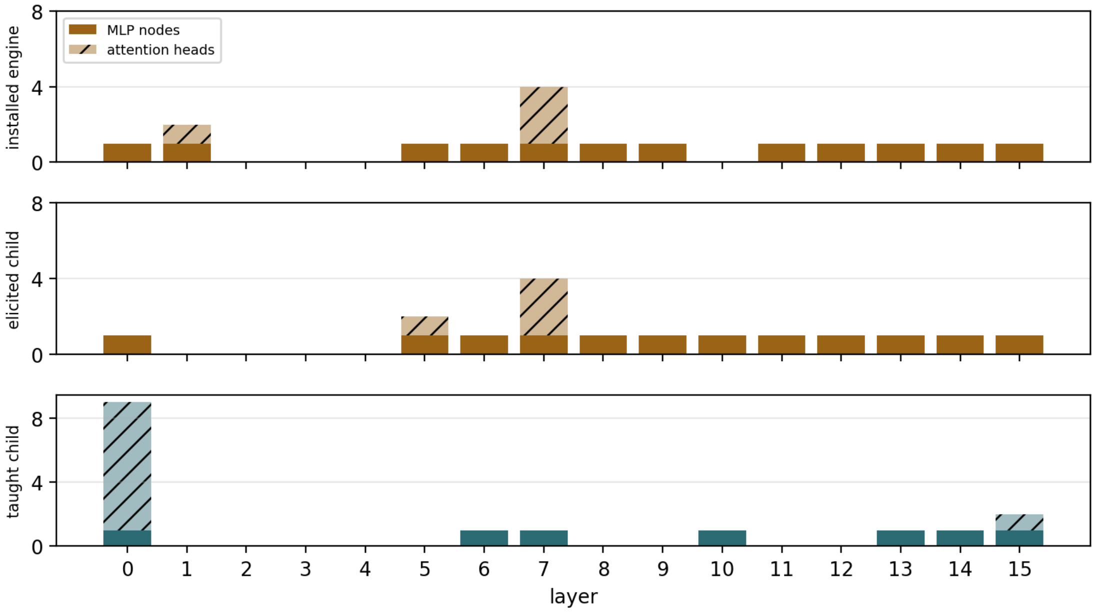

Figure 2: Top-16 node layer histograms (solid = MLP, hatched = attention) for the installed engine, the elicited child, and the taught child.

C. Dynamics: how the machinery comes into being

R5 The elicited circuit exists before training; the taught circuit is built mid-run — through the EDL hump.

Metric. Jaccard@32 of each training snapshot's circuit against the run's final circuit.

Result. Figure 3. Elicit: 0.488 at step 1 (16× chance), a brief interface-wiring transient (dip to ∼0.42 with a strong map at step 154), locked from ∼step 1.5K of 11.5K (0.561; 82% of ceiling, 97% of the J@64 ceiling by 8.4K). Teach: noise floor (≈0.22–0.28) until ∼step 1.3K, then 0.27 → 0.44 over steps 1.3K–5.4K — the region where its EDL peaks.

Elicit vs. teach. Elicitation finds and wires; teaching constructs. The elicited circuit finishes stabilising before the taught one begins to exist, and the teaching hump of R1 is literally the circuit being assembled.

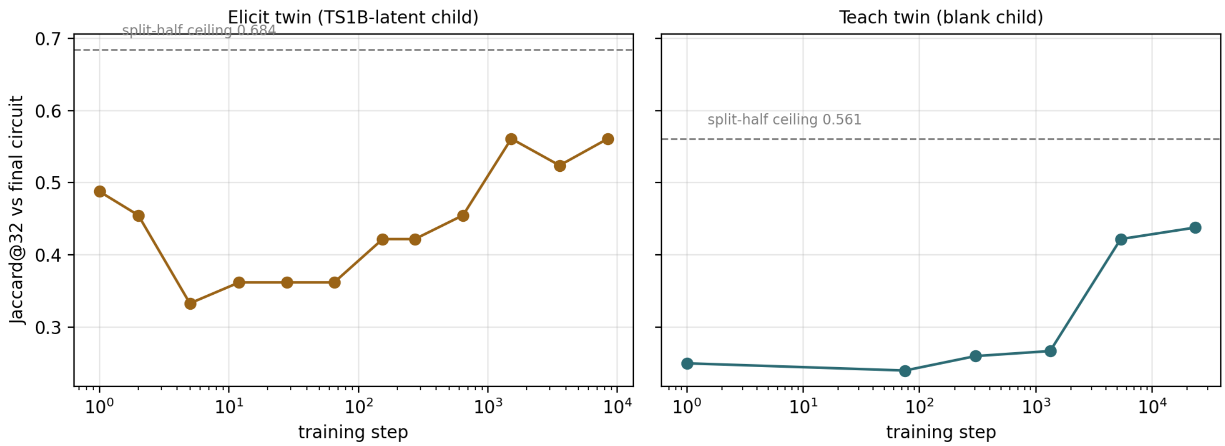

Figure 3: Circuit formation, elicit (left) vs teach (right); dashed = split-half ceiling.

R6 Elicitation's gradient dies; teaching's grows.

Metric. Pre-clip global gradient norm at every update, from the runs' `gradstats` logs: peak, mean over first 1% / middle / last 10% of steps, cumulative gradient mass $\sum_t \|g_t\|$, and its split across weight classes.

Result. Table 5, Figure 4. Both arms start at the same scale (first-1% mean 0.18 vs 0.17). Elicit then decays ×3.3 to a floor of 0.056 (peak 0.44 at step 96). Teach *grows* ∼40× to 6.8 (peak 17.4 at step 23,657 — still rising when the run stopped). Cumulative gradient mass 783 vs 116,877 (∼150×). Across n the arms are indistinguishable up to 10K and diverge from 31.6K on (mass ratio 7× → 55× → 95× → 150×). Teach's gradient mass migrates out of the MLPs (share 0.47 → 0.01) into attention (VO 0.87).

Elicit vs. teach. The optimizer's view of R5: eliciting converges and its gradient dies; teaching keeps generating gradient pressure that grows with the structure being built, aimed at attention — the layer-0 army of R4 being written. This is the clearest dynamical discriminator, and it sets up R7: what all that gradient actually bought.

R7 Teaching writes more into the weights and buys far less per unit written.

Metric. Relative write $\|\Delta W\|/\|W\|$ at n=1M (exact from LoRA factors); total travel $\|\Delta W_t\|_F$ from adapter snapshots; capability per unit travel = final EM / total travel.

1M fine-tune	elicit	teach
peak grad norm (step)	0.441 (96)	17.4 (23,657)
mean norm: first 1% / middle / last 10%	0.183 / 0.060 / 0.056	0.169 / 5.21 / 6.80
first 1% ÷ last 10%	×3.3 (decays)	×0.02 (grows ∼40×)
cumulative gradient mass $\sum_t \lVert g_t \rVert$	783	116,877
gradient mass per step	0.068	4.68
mass share QK / VO / MLP	.18 / .55 / .27	.12 / .87 / .01

Table 5: Raw gradient strength, 1M fine-tunes (pre-clip global norm at every update).

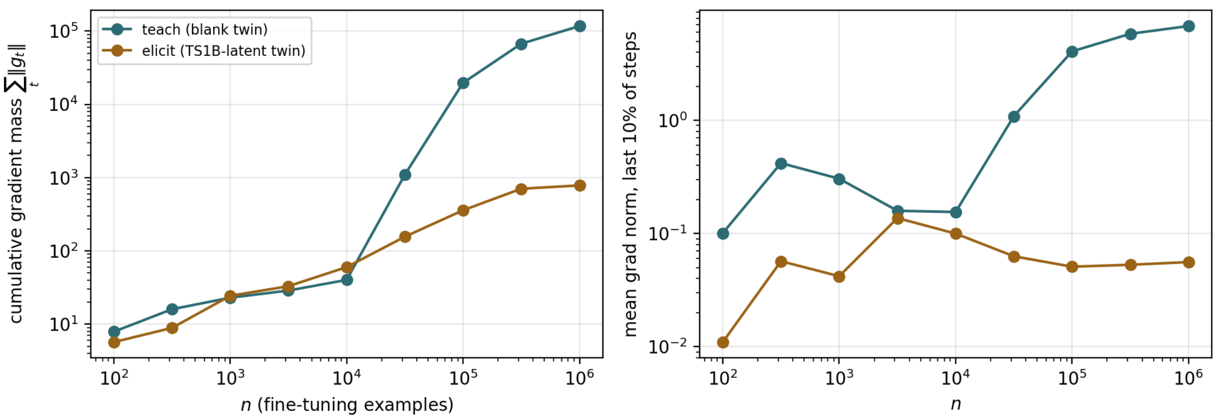

Figure 4: Cumulative gradient mass (left) and late-run gradient norm (right) vs n; teal = teach, gold = elicit.

Result. Table 6. Relative write 0.212 (teach) vs 0.095 (elicit), 2.2×; total travel 457 vs 120 (3.8×) over 2× as many steps (23.5K vs 11.5K); capability per unit travel 2.0×10^{-4} vs 8.1×10^{-3}, ∼40×. The travel *profiles* of the two arms coincide (both burst then decay): AdamW's $\sqrt{v}$-normalisation compresses R6's 150× gradient-mass gap into a 3.8× step-size gap with a shared shape.

Elicit vs. teach. What R6's gradients bought: teaching writes 2–4× more into the weights yet buys ∼40× less capability per unit written — eliciting *connects* what exists, teaching *writes* what does not. Update size is the muted echo of the gradient signal; magnitude and efficiency, not timing, are where the weights show the regime.

	elicit (TS1B-latent)	teach (blank)
relative write $\lVert \Delta W \rVert / \lVert W \rVert$ (1M)	0.095	0.212
steps to convergence	11,500	23,496
peak / final writing speed (per step)	0.218 / 0.018	0.306 / 0.022
total weight travel $\lVert \Delta W \rVert_F$	120	457
final exact match	0.981	0.093
capability per unit travel	8.1×10^{-3}	2.0×10^{-4}

Table 6: Weight-space write, elicit vs teach (travel curves in `05_weights/fig_weight_travel`).

R8 Teaching writes a loud new output channel; elicitation's shift is the answer itself.

Metric. Residual-stream shift between parent and child on identical inputs, $\|h_\ell^{\text{child}} - h_\ell^{\text{parent}}\| / \|h_\ell^{\text{parent}}\|$ per layer (both models in fp32), at the answer position of 256 NL problems and on 256 held-out TinyStories passages; direction statistics of the 256 per-problem shift vectors — energy fraction of the top singular direction (PC1) and cosine to the mean — each next to the same statistic on the parent's own states at those positions (the anisotropy reference).

Result. Figure 5, Table 7. Final-layer shift at the answer position: teach **14.1**× the parent's norm (layers 13/14: 3.2×/5.0×) vs elicit **1.7**×, built gradually from layer 9; over every prompt position 5.6× vs 1.0×. Shift on generic text: 0.156 vs 0.160 — identical. Direction at the read-out: elicit PC1 0.063, cosine-to-mean 0.26 (parent states: 0.23 / 0.51) — no dominant shared direction; teach PC1 0.35 — the larger common component.

Elicit vs. teach. R6's 150× gradient mass and R7's 3.8× travel end up as an 8× louder write into the final residual stream, for a model 10× worse: a new, large-norm output channel. The elicited child differs from its parent by a *per-problem* vector — the answer routed to the output — not by a reusable "answer-mode" switch; that is exactly why the mean-vector patch in R11 yields format only while per-prompt states unlock the task. Elicit vs. teach: elicitation adds content to an existing state; teaching adds a loud channel. (The proposed surgical-vs-diffuse reading of this metric did not hold; see the non-discriminators.)

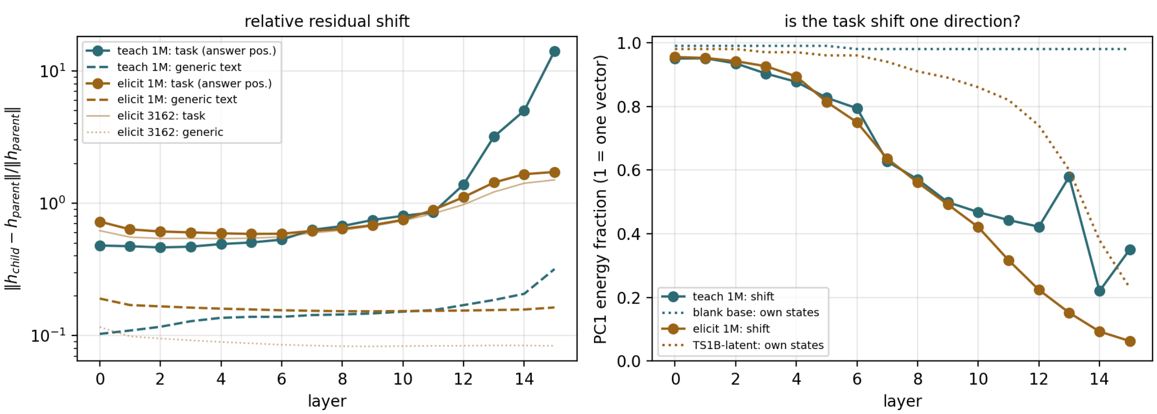

Figure 5: Left: relative residual shift by layer at the answer position (solid) and on generic text (dashed). Right: PC1 energy fraction of the per-problem shift vectors, with the parent-state reference dotted.

D. The latent state: what it means and what it permits before training

R9 In the latent parent, the words already reach the engine.

Metric. Cross-format activation matching: per node, mean cosine between the activation on the NL rendering and on the symbol rendering of the *same* problem, minus the mismatched-pair baseline ("does `What is the sum of 621 and 5068?` drive the node into the same problem-specific state as `621 + 5068 = ?`").

Result. Engine without the binding dose: MLP index ≈ 0.00–0.01 at every layer. TS1B-latent (binding installed): +0.07–0.13 across layers 7–15, ∼10×. Attention-head indices are

	elicit 1M	elicit 3162	teach 1M	construction (full FT)
shift at answer position, final layer	1.72	1.50	**14.13**	1.10
shift at answer position, layer mean	0.87	0.78	1.93	0.61
shift over all prompt positions, final layer	0.96	0.82	5.55	0.88
shift on generic text, layer mean	0.160	0.089	0.156	1.118
PC1 of shift, final layer (parent states)	**0.063** (0.23)	0.093 (0.23)	0.351 (0.98)	0.603 (0.98)
cosine-to-mean, final layer (parent states)	0.26 (0.51)	0.32 (0.51)	0.33 (0.99)	0.79 (0.99)
final exact match	0.981	0.541	0.093	–

Table 7: Residual-stream shift parent→child (N=256; generic text = 256 held-out TinyStories passages, 64 tokens). "Construction" is the full fine-tune that built TS1B-latent from the blank base, shown as an anchor.

confounded by shared digit tokens and are excluded. The geometry of the answer-position state agrees (final layer, 256 problems): the latent parent's states have cosine-to-mean 0.51 and PC1 0.23 — problem-specific — while the blank base's have 0.99 and 0.98 — problem-blind.

Elicit vs. teach. This is what "latent" means internally: before any target training, NL input already activates the arithmetic MLPs of the elicit parent, while the teach parent has no arithmetic MLPs to activate. It also confirms the binding dose installed an access route, not the task.

R10 The latent parent already holds the answer in its verbalizable space; the blank base holds nothing. Elicitation forms the answer where the engine forms it; teaching forms it only at the last layer.

Metric. Logit lens, Jacobian lens (J-lens: the residual at layer ℓ transported to the final-layer basis by the corpus-average Jacobian $J_\ell = \mathbb{E}[\partial h_{\text{final}}/\partial h_\ell]$ over held-out stories, then unembedded) and R-lens (the J-lens with a relevance-propagation backward pass), at the answer position of 256 problems. Scored token = first digit chunk of the answer (sign appended to the prompt for negative answers). Read-outs: median rank of the answer token (chance 64K of 128K), logit-diff against a mismatched problem's answer, settled depth (first layer from which the token stays top-1).

Result. Figure 6, Table 8. Latent parent on the NL target, J-space rank of the answer: 2,606 at layer 12, 510 at layer 14, 174 in the output distribution — while it emits a rewrite; logit-diff rising from layer 8. Blank base: 55–80K at every layer, logit-diff 0.00 everywhere, in all three lenses. Elicited child: rank 139 / 6 / 1 at layers 12 / 13 / 14, settled median layer 14 — the same trajectory as the parent's engine on symbols (122 / 6 / 1) and the same onset as the parent on NL (layer 8). Taught child: rank 41,677 / 19,933 / 8 at layers 12 / 14 / 15; top-1 0.02 → 0.08 → 0.11 over the last three layers.

Elicit vs. teach. "Latent" in the workspace sense: the answer is present and rising in the parent's verbalizable space before any target training, and elicitation amplifies that late read-out without moving it — parent and child lens trajectories coincide layer by layer on the engine's own surface. Teaching has no signal to amplify: its answer exists nowhere in the stack until the last layer writes it (the loud final write of R8). Elicit vs. teach: depth is inherited from the engine under elicitation and pinned to the last layer under teaching. The "middle layers" form of the prediction does not hold — formation is at layers 13–14 of 16 in parent and elicited child alike.

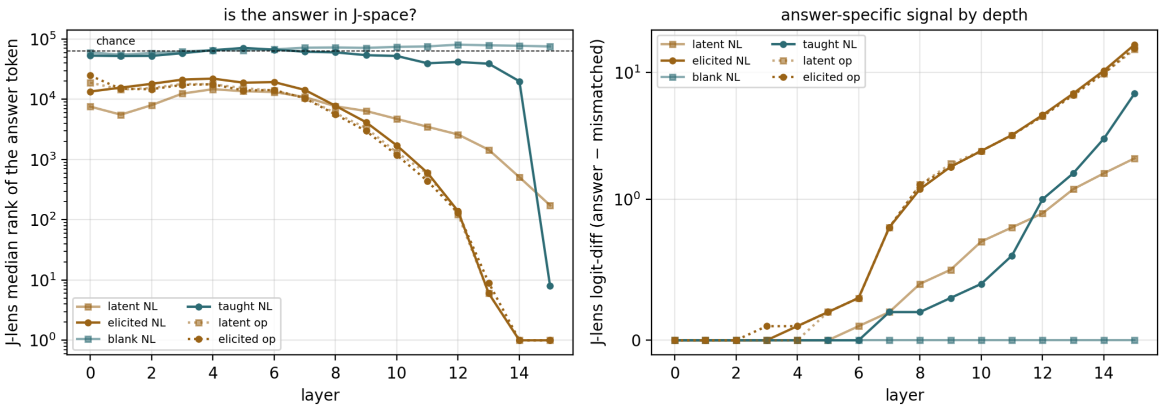

Figure 6: J-lens read-out of the answer token by layer: median rank (left, chance 64K) and logit-diff (right). Squares = parents, circles = 1M children; dotted = symbol surface.

model	surface	own acc	rank L12 (logit / J / R)	rank L14 (logit / J / R)	read-out
TS1B-latent (parent)	NL	0.047	5,704 / 2,606 / 2,730	436 / 510 / 424	174
elicited child	NL	0.957	1,584 / 139 / 464	1 / 1 / 1	1
blank base	NL	0.000	76,049 / 80,840 / 78,243	77,707 / 77,320 / 79,116	75,441
taught child	NL	0.109	51,167 / 41,677 / 44,335	35,620 / 19,933 / 21,156	8
TS1B-latent (parent)	symbols	0.836	1,224 / 122 / 334	1 / 1 / 1	1
elicited child	symbols	0.855	1,360 / 131 / 399	1 / 1 / 1	1

Table 8: Lens depth of the answer's first digit token: median rank among 128K entries (chance 64K) under the logit lens / J-lens / R-lens; "own acc" = the model's own first-digit-token accuracy on the 256 held-out problems; read-out = last layer, where all lenses coincide with the model's output.

R11 The latent capability can be reached with frozen weights; the absent one cannot.

Metric. Exact match on the target task under zero-training interventions: (a) self-chain — the model rewrites the question itself, then computes its own rewrite; (b) donor patching — the fine-tuned child's activations at the 32 circuit nodes written into the parent (mean vector, or per-prompt states); random-node controls.

Result. Figure 7, Table 9. Elicit parent: direct 0.000 → self-chain **0.328** (= 0.484 × 0.668; the composition law parse×compute held on four model variants); per-prompt states 0.125 (random nodes: 0.000); mean vector: format only. Teach side: blank + same doses 0.000; blank + chain 0.000 (a parser without an engine); taught donor into blank twin 0.000 in every condition.

Elicit vs. teach. Elicit = one composition step or one state-patch away from the target; teach = nothing underneath to switch on. The teach-side nulls double as the leakage control for the whole construction — and they close the loop with R2: the machinery the patch activates is the same installed engine the fine-tune later reuses.

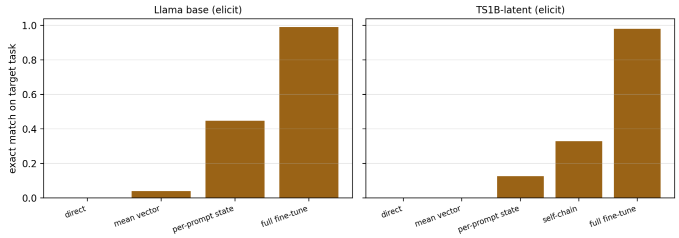

Figure 7: Intervention ladder on the elicit parent (weights frozen except the last bar).

Patient + intervention (k=32 circuit nodes)	condition	format	EM
TS1B-latent + child mean shift	circuit nodes	0.164	0.000
TS1B-latent + child per-prompt states	circuit nodes	0.359	**0.125**
	random-32 nodes	0.051	0.000
TS1B-latent self-chain (no donor)	own rewrite + "= "	–	**0.328**
blank + binding dose, self-chain	parser without engine	–	0.000
blank + taught donor	every condition	0.000	0.000

Table 9: Zero-training interventions, TinyStories twins.

Validation results (not regime differences, but what licenses R2–R11)

- **Both circuits are real and load-bearing.** True activation patching: elicited child top-8 of 528 nodes = 0.997 sufficient / 0.998 necessary (top-32: 0.999 / 1.000); taught child top-8 necessity 0.986, top-32 ∼0.999. Every overlap above compares provably load-bearing sets.
- **Lens depth is scored on the first digit token.** Scoring the raw first token credited the taught child with layer-5 formation on 48 problems — all subtractions with a negative answer, whose first token is the minus sign, produced by a comparison circuit. With the sign appended to the prompt the effect vanishes (first-token accuracy 0.31 → 0.11; no early route); the other three models were unaffected.
- **Construction controls.** Identical doses on the blank twin change nothing on the target (0.000 in every probe); a sum-only binding dose leaves "difference" unbound (per-word binding); the composition law chain = parse × compute held on four variants.

Informative non-discriminators (measured, reported for completeness)

- Effective rank of ΔW (participation ratio): elicit 7.1, teach 5.2 of 512 — both energy-concentrated; rank does not separate the regimes. Alignment of ΔW with the base's top-64 singular directions is near the 0.058 baseline for both.
- Temporal profile of weight travel: burst-then-decay in both arms (R7) — an effect of AdamW normalisation; the underlying gradient dynamics do discriminate (R6).
- Prompt brittleness (16-shot collapse) is a property of this lineage's training-sequence structure, not of the regime: both children score ∼0 at 16 shots.
- Generic-text displacement and depth spread of the residual shift (metric 10): the two 1M children move held-out stories by the same relative amount (0.160 vs 0.156) and both write

the task shift in the last three layers (82% / 99.6% of increment energy). The task/generic ratio is even reversed (5.4 elicit vs 12.4 teach) because of the taught child's loud final write (R8). The full-fine-tuned construction displaced generic text 7× more (1.12), but confounds full FT with regime.

- Edge-level ΔS within the elicit arm: the edge instrument's split-half noise floor (0.61–0.62) exceeds the measured churn on both eval surfaces — a sensitivity limit.